

Subject Description Form

Please read the notes at the end of the table carefully before completing the form.

Subject Code	SFT545
Subject Title	Advanced Fiber Materials
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	N/A
Objectives	The subject is designed to provide students with in-depth knowledge of fibre materials and processing for wearable technology, including fibre and fabric development, as well as colouration and finishing processing. It will provide up-do-date development information on the functions and applications of advanced fibrous materials for intelligent wearables as well as the latest processing technologies. It will also introduce methods for evaluating the performance of fibrous materials.
Intended Learning Outcomes <i>(Note 1)</i>	Upon completion of this subject, students will be able to: 1. Understand the interrelationships between fibers, yarns, and fabric structures in textiles and apparel materials that can be used in intelligent wearables, as well as the principles of colouration and finishing technologies. 2. Comprehend and describe the various quality evaluation and process control systems in modern textiles and apparel manufacturing. 3. Communicate with professionals on technical and economic knowledge applicable to actual industrial situations.
Subject Synopsis/ Indicative Syllabus <i>(Note 2)</i>	1. Fibers classification and properties 2. Yarns structure and properties and processing technology. 3. Fabrics structure and properties and processing technology. 4. Textile colouration and finishing technology. 5. Quality evaluation and/or process control of fibrous materials (fibres, yarns and fabrics), and coloration-finishing technology 6. Application of fibrous materials

	7. Functional fibrous materials and their development																										
Teaching/Learning Methodology <i>(Note 3)</i>	The subject consists of lectures, seminars and tutorials. The lectures will cover the key topics and knowledge delivery. Students are required to do investigations on new materials and processes, as well as potential developments and applications, and then give presentations. Students are encouraged to participate in seminar and tutorial discussions.																										
Assessment Methods in Alignment with Intended Learning Outcomes <i>(Note 4)</i>	<table border="1"> <tr> <th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="3">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr> <tr> <th>1</th><th>2</th><th>3</th></tr> <tr> <td>1. Coursework</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td></tr> <tr> <td>2. Examination</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td></tr> <tr> <td>Total</td><td>100 %</td><td colspan="3"></td></tr> </table> In the continuous assessment (coursework), we focus on training students: self-learning ability to acquire technical knowledge and improve communication skills. By assigning homework assignments and topics to students, and asking students to answer, students can obtain self-learning ability and enhance professional knowledge through search and selection. Students can improve communication skills by organizing group projects and requiring interaction between group members, as well as asking groups to submit reports and give presentations in class. Through the examination, students' academic performances are assessed by examining how well they master concept definitions, explain principles, describe and analyze application examples, and provide rationality for different fibrous materials, processing technologies and quality issues. The materials submitted for this assessment must be the student's own work. The submitted work may not be accepted for the purpose of assessment if its authenticity is questionable. Submitting GenAI-generated materials as students' own work or part of their work is an act of academic dishonesty. Students who are found committing academic dishonesty will face disciplinary actions.				Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			1	2	3	1. Coursework	50%	✓	✓	✓	2. Examination	50%	✓	✓	✓	Total	100 %			
Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)																									
		1	2	3																							
1. Coursework	50%	✓	✓	✓																							
2. Examination	50%	✓	✓	✓																							
Total	100 %																										
Student Study Effort	Class contact:																										

Expected	▪ Lecture	27 Hrs.
	▪ Tutorials/Seminars	12 Hrs.
	Other student study effort:	
	▪ Preparation for presentation	36 Hrs.
	▪ Assignments	30 Hrs.
	Total student study effort	105Hrs.
Reading List and References	<u>Selected Reading Books</u> 1. Xiaoming Tao, <u>Smart Fibres, Fabrics and Clothing</u>, Cambridge, England: Woodhead Pub., 2001. 2. Bernard P. Corbman, <u>Textiles: Fiber to Fabric</u>, Gregg Division McGraw-Hill ; 6th ed, 1985. 3. Boca Raton, Horrocks, A. Richard and Anand, Subhash, <u>Handbook of Technical Textiles</u>, Cambridge, England: Woodhead Pub., CRC Press, 2000. 4. Carl A. Lawrence, <u>High performance textiles and their applications</u>, Cambridge, Woodhead Publishing, 2014. 5. Morton, W.E., Hearle, J.W.S., <u>Physical Properties of Textile Fibres</u> 3rd ed., Butterworths, 1992. 6. Textile Institute, <u>Textile Terms and Definition</u>, 9th ed., 1991. 7. Miller E., <u>Textile Properties and Behaviour in Clothing use</u>, Batsford Technical, 1984. 8. Carr, H., Carr and Latham's, <u>Technology of Clothing Manufacture</u>, 4th ed., Oxford: Blackwell Publishing, 2008. 9. B P Saville, <u>Physical Testing of Textile</u>, Woodhead Pub., CRC Press, 1999. 10. Xiaoming Tao, <u>Handbook of Smart Textiles</u>, Springer, 2015. 11. <u>Process control in textile manufacturing</u>, Woodhead Publishing Limited, 2013. 12. <u>Advances in the dyeing and finishing of technical textiles</u>, Woodhead Publishing Limited, 2013. 13. <u>Chemical finishing of Textiles</u>, Woodhead Publishing Limited, 2004. <u>Supplementary Reading Journals</u> 1. High Performance Textiles 2. Textile Research Journal <u>Websites</u> www.merchandisers.hk	

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Note 1: Intended Learning Outcomes

Intended learning outcomes should state what students should be able to do or attain upon subject completion. Subject outcomes are expected to contribute to the attainment of the overall programme outcomes.

Note 2: Subject Synopsis/Indicative Syllabus

The syllabus should adequately address the intended learning outcomes. At the same time, overcrowding of the syllabus should be avoided.

Note 3: Teaching/Learning Methodology

This section should include a brief description of the teaching and learning methods to be employed to facilitate learning, and a justification of how the methods are aligned with the intended learning outcomes of the subject.

Note 4: Assessment Method

This section should include the assessment method(s) to be used and its relative weighting, and indicate which of the subject intended learning outcomes that each method is intended to assess. It should also provide a brief explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes.